



**ITM SKILLS
UNIVERSITY**

Case Study Report: CartMax E-Commerce Automated Checkout Coupon Match System

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Data Structures and Algorithms Project

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1. Introduction to the Case Study

Modern e-commerce platforms such as Amazon, Flipkart, and Myntra process millions of customer transactions daily. During sales and promotional

events, thousands of users simultaneously search for discount coupons, add products to their carts, and proceed to checkout. Managing these operations efficiently requires the use of optimized data structures and algorithms.

This case study focuses on the design and implementation of CartMax, an automated e-commerce checkout system developed using C++. The system leverages advanced Data Structures and Algorithms (DSA) to provide fast coupon lookup, cart management, membership verification, inventory reservation, product recommendations, checkout processing, and discount optimization.

The project demonstrates how different data structures can be applied to solve real-world challenges in modern online shopping platforms while maintaining high performance and scalability.

2. Problem Statement / Case Background (Abstract)

Background

E-commerce platforms often face several operational challenges during high-traffic periods:

- Slow coupon code searches as users type promotional codes.

- **Difficulty reverting cart changes after removing items or discounts.**
- **Checkout congestion when thousands of customers attempt payment simultaneously.**
- **Inventory overselling due to delayed stock reservations.**
- **Lack of intelligent product recommendations.**
- **Complex coupon stacking scenarios that fail to provide maximum customer savings.**

Abstract

This case study presents the design of CartMax, a menu-driven e-commerce management system that maps specific business problems to appropriate data structures and algorithms.

The system features:

- **A Promo Code Suggestion Engine for instant coupon lookup.**
- **A Cart Undo System for reverting pricing changes.**
- **A Checkout Queue Manager for processing customers in FIFO order.**
- **A Membership Verification Module for instant customer lookup.**
- **A Cart Monitoring System to identify high-value orders.**
- **A Product Recommendation Network for bundle suggestions.**
- **A Discount Optimization Engine using Dynamic Programming.**

- **A Stock Reservation System to prevent inventory overselling.**

The application demonstrates how enterprise-level e-commerce operations can be simulated efficiently using core DSA concepts.

3. Case Study Design

The CartMax system is designed as a modular application consisting of the following components:

1. Promo Code Lookup System

Uses a Trie Data Structure to store and search coupon codes efficiently.

Benefits:

- **Fast prefix-based searches.**
- **Instant coupon suggestions.**
- **$O(L)$ lookup complexity where L is coupon length.**

2. Cart Price Undo System

Uses a Stack (LIFO) to maintain cart price history.

Benefits:

- **Instant rollback of cart changes.**
- **$O(1)$ undo operation.**

- Supports cancellation of discounts and item removals.

3. Checkout Processing System

Uses a Queue (FIFO) to process customers in the exact order they entered checkout.

Benefits:

- Fair customer processing.
- Simulates real-world payment queues.
- $O(1)$ insertion and deletion.

4. Membership Verification

Uses a Hash Map for customer membership storage.

Benefits:

- $O(1)$ average lookup time.
- Fast verification of Premium and Gold members.

5. Cart Monitoring System

Uses a Priority Queue (Max Heap) to identify the highest-value shopping carts.

Benefits:

- Fraud monitoring.
- High-value customer tracking.
- $O(\log N)$ insertion.

6. Product Recommendation Engine

Uses a Graph Data Structure to model relationships between products frequently purchased together.

Benefits:

- **Bundle suggestions.**
- **Cross-selling opportunities.**
- **Personalized shopping experience.**

7. Discount Optimization Engine

Uses Dynamic Programming (Bitmask DP) to determine the optimal sequence of coupon application.

Benefits:

- **Maximum customer savings.**
- **Efficient coupon stacking.**
- **Eliminates redundant calculations.**

8. Inventory Reservation System

Uses Hash Maps to manage product stock and reserve items immediately when added to the cart.

Benefits:

- **Prevents overselling.**
- **Real-time inventory updates.**
- **$O(1)$ stock verification.**

4. Methods and Algorithms Applied

Key Methods and Logic :

1. Fast Coupon Search

The system stores coupon codes in a Trie and generates suggestions based on user-entered prefixes.

2. Undo Mechanism

Stack-based history tracking allows instant reversal of cart modifications.

3. Checkout Scheduling

Queue operations ensure customers are processed fairly and sequentially.

4. Product Relationship Mapping

Graph structures represent connections between products to generate recommendations.

5. Dynamic Programming Optimization

Bitmask Dynamic Programming evaluates all valid coupon combinations and identifies the sequence that provides the maximum discount.

6. Inventory Control

Hash Maps enable real-time stock tracking and item reservation.

Technology Stack Used

Programming Language

- **C++**

Concepts Applied

- **Object-Oriented Programming (OOP)**
- **Data Structures and Algorithms**
- **Dynamic Programming**
- **Graph Traversal**
- **Heap Operations**
- **Trie-Based Searching**

Development Environment

- **Visual Studio Code**
- **Git & GitHub**

5. Results and Conclusion

Findings

- **Trie-based coupon lookup significantly improved promo code search performance.**
- **Stack operations enabled instant cart rollback with $O(1)$ complexity.**
- **Queue implementation ensured orderly checkout processing during high traffic.**
- **Hash Maps provided near-instant membership and inventory lookups.**
- **Priority Queues efficiently identified high-value carts for monitoring.**
- **Graph-based recommendations improved product discovery.**

- **Dynamic Programming successfully determined optimal coupon stacking strategies.**
- **Inventory reservation eliminated overselling scenarios.**

Conclusion

The CartMax E-Commerce Automated Checkout Coupon Match System demonstrates the practical implementation of multiple data structures and algorithms in a real-world business scenario. By integrating Trie, Stack, Queue, Hash Map, Priority Queue, Graph, and Dynamic Programming, the system efficiently handles coupon management, inventory control, customer processing, and discount optimization.

The project successfully illustrates how fundamental DSA concepts can be applied to solve complex challenges faced by modern e-commerce platforms such as Amazon, Flipkart, and Myntra.

2.10 Screenshots

Figure 1: CartMax Main Menu Interface

```
===== CARTMAX =====  
1  View Products  
2  Add To Cart  
3  Remove From Cart  
4  View Cart  
5  Undo Last Change  
6  Promo Suggestions  
7  Membership Check  
8  Product Recommendations  
9  Add Customer To Checkout  
10 View Checkout Queue  
11 Process Checkout  
12 Best Discount  
13 Highest Value Cart  
14 Top 3 Carts  
15 Order History  
0  Exit
```

Figure 2: Product Listing Output

```
===== PRODUCTS =====  
P4 | Headphones | Rs.3500 | Stock: 20  
P2 | Mouse | Rs.1000 | Stock: 50  
P3 | Keyboard | Rs.2500 | Stock: 25  
P1 | Laptop | Rs.50000 | Stock: 10
```